

VICTORIA BEACH, MB, Canada

WMO: 715520

Lat: 50.6944N

Lon: 96.5625W

Elev: 220

StdP: 98.71

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.3	-28.0	-33.7	0.2	-30.2	-31.3	0.2	-27.9	14.2	-12.5	13.0	-13.3	3.1	320	0.730

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	28.0	21.6	26.6	20.8	25.2	20.1	23.0	26.4	21.9	25.3	20.9	24.2	4.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	17.0	25.0	20.7	15.8	24.1	19.7	14.8	23.2	69.3	26.5	65.1	25.1	61.6	24.3	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
13.9	12.2	10.9	DB	-33.6	32.7	2.1	2.3	-35.1	34.3	-36.3	35.7	-37.5	37.0	-39.1	38.7	
			WB	-33.4	24.9	2.2	1.2	-35.0	25.8	-36.3	26.5	-37.5	27.2	-39.1	28.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-15.3	-14.0	-6.5	2.4	10.2	17.4	21.0	20.4	14.9	6.9	-2.5	-11.9	(6)
(7)		DBStd	14.08	7.66	7.57	7.52	5.41	4.92	3.55	2.81	3.07	4.00	4.36	5.92	7.33	(7)
(8)		HDD10.0	3438	785	673	512	231	59	1	0	0	7	116	376	679	(8)
(9)		HDD18.3	5567	1043	906	770	478	256	58	9	14	116	353	626	938	(9)
(10)		CDD10.0	1126	0	0	0	3	64	223	339	321	153	21	1	0	(10)
(11)		CDD18.3	213	0	0	0	0	2	30	91	77	13	0	0	0	(11)
(12)		CDH23.3	797	0	0	0	0	10	100	351	303	32	1	0	0	(12)
(13)		CDH26.7	131	0	0	0	0	1	14	54	56	5	0	0	0	(13)
(14)	Wind	WSAvg	5.2	5.0	4.6	4.8	4.7	5.0	4.9	5.0	5.4	5.9	6.3	6.0	5.1	(14)
(15)	Precipitation	PrecAvg	545	22	15	24	28	64	92	79	74	59	42	28	23	(15)
(16)		PrecMax	751	59	29	75	62	120	195	170	148	117	91	85	55	(16)
(17)		PrecMin	358	7	1	3	10	19	38	18	16	19	9	5	6	(17)
(18)		PrecStd	96	12	8	15	12	28	42	31	31	26	23	16	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	3.7	10.5	17.1	24.4	28.6	30.3	30.7	26.6	20.6	12.3	3.2	(19)
(20)			MCWB	1.2	2.2	7.1	9.6	15.7	21.6	23.1	22.2	19.9	14.9	9.1	1.4	(20)
(21)		2%	DB	0.3	1.7	6.7	14.0	21.3	25.9	28.0	28.1	23.8	17.2	8.6	1.1	(21)
(22)			MCWB	-0.5	0.7	3.9	7.9	14.3	19.5	22.0	21.3	18.6	13.1	6.3	0.1	(22)
(23)		5%	DB	-2.0	0.2	4.6	11.6	18.9	24.1	26.6	26.4	22.0	14.9	6.9	-0.3	(23)
(24)			MCWB	-2.6	-0.6	2.6	6.9	13.4	18.7	21.4	20.5	17.3	11.9	5.1	-0.9	(24)
(25)		10%	DB	-4.3	-2.6	2.8	9.4	16.9	22.6	25.3	24.9	20.3	13.0	4.7	-1.9	(25)
(26)			MCWB	-4.8	-3.4	1.4	5.7	12.3	17.8	20.7	19.9	16.6	10.4	3.5	-2.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	2.5	7.5	10.2	17.6	23.1	24.9	24.2	21.3	15.5	9.4	1.7	(27)
(28)			MCDWB	2.1	3.5	10.2	15.0	21.4	26.6	27.9	27.8	24.5	19.0	11.9	3.0	(28)
(29)		2%	WB	-0.6	0.9	4.3	8.7	15.8	20.9	23.5	22.5	19.6	13.8	6.8	0.3	(29)
(30)			MCDWB	0.2	1.4	6.3	12.5	19.5	24.2	26.6	26.4	22.6	16.6	8.2	1.1	(30)
(31)		5%	WB	-2.6	-0.6	2.7	7.3	14.3	19.6	22.4	21.4	18.3	12.1	5.1	-0.9	(31)
(32)			MCDWB	-2.0	0.2	4.4	11.3	17.7	22.9	25.5	25.2	21.1	14.4	6.6	-0.3	(32)
(33)		10%	WB	-4.8	-3.3	1.5	5.8	13.0	18.5	21.4	20.5	17.1	10.3	3.4	-2.5	(33)
(34)			MCDWB	-4.3	-2.6	2.7	9.2	16.0	21.7	24.4	24.1	19.8	12.8	4.6	-2.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	9.5	9.9	9.2	8.3	7.3	7.1	7.3	6.5	5.2	4.9	6.8	(35)
(36)			MCDBR	8.4	7.5	9.0	11.4	11.0	9.6	8.8	9.7	9.0	8.5	6.6	7.1	(36)
(37)			MCWBR	7.9	6.5	6.8	6.5	5.7	4.8	4.6	4.7	5.0	5.2	4.8	6.7	(37)
(38)		5% WB	MCDWB	8.1	7.4	8.3	10.7	9.2	8.1	7.9	8.7	8.0	8.3	6.2	7.4	(38)
(39)			MCWBR	7.6	6.4	6.4	6.4	5.4	5.0	4.9	5.0	5.1	5.3	4.6	6.9	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.265	0.288	0.295	0.348	0.361	0.381	0.390	0.374	0.344	0.300	0.272	0.248	(41)	
(42)		taud	2.223	2.174	2.323	2.361	2.404	2.371	2.373	2.425	2.494	2.574	2.436	2.289	(42)	
(43)		Ebn at Noon	762	837	911	894	895	875	861	858	851	839	769	735	(43)	
		Edh at Noon	70	97	101	111	111	116	114	103	86	66	57	56		
(44)	All-Sky Solar Radiation	RadAvg	1.21	2.37	3.66	5.03	5.46	5.79	6.02	5.00	3.54	2.05	1.10	0.88	(44)	
(45)		RadStd	0.12	0.17	0.27	0.34	0.46	0.46	0.28	0.35	0.32	0.26	0.15	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.78	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	-0.59	-0.79	N/A	N/A	N/A	N/A

Nomenclature: See separate page